JENNIFER JAHNCKE, PHD

Portland, OR

EDUCATION

2024 | 2018

Oregon Health & Science University

PhD in Neuroscience — Portland, OR

- Advisor: Kevin M. Wright, PhD, Vollum Institute, OHSU
- Dissertation: A functional role for Dystroglycan at inhibitory synapses across multiple brain regions

2014 | 2010

University of California, Davis

BS in Psychology, Minor in Neuroscience — Davis, CA

Selected Courses

- Analytical and Quantitative Light Microscopy, MBL, 2024, Woods Hole, MA
- Behavioral Neuroscience Responsible Conduct of Research, OHSU, 2024, Portland, OR
- Neurohackademy, UW, 2023, Seattle, WA
- Programming for Biology, CSHL, 2022, Cold Spring Harbor, NY
- Biostatistics, OHSU, 2020, Portland, OR
- Data Visualization, OHSU, 2020, Portland, OR
- Practice and Ethics of Science, OHSU, 2018, Portland, OR
- Light and Fluorescence Microscopy, UC Davis, 2017, Davis, CA

RESEARCH EXPERIENCE

Present | 2024

Postdoctoral Researcher

Vollum Institute, OHSU — Portland, OR

- Molecular and morphological characterization of a neural circuit lacking two postsynaptic cell adhesion molecules, resulting in a complete loss of a population of synapses.
- Identification and characterization of an evolutionarily conserved novel architecture of cerebellar GABAergic synapses.
- Development and initial validation of gene therapy to alleviate neurological deficits in mouse models of dystroglycanopathy.

2024 | 2018

Graduate Student Researcher

Neuroscience Graduate Program, OHSU — Portland, OR

- Functional characterization of the scaffolding protein Dystroglycan at inhibitory synapses across the brain in mouse models of dystroglycanopathy.
- Identified novel interacting partners of Dystroglycan in the central nervous system.

CONTACT INFO

jahncke@ohsu.edu
+1 925-895-2421
jennifer-jahncke.netlify.app
github.com/jnjahncke
0000-0003-2319-6109

SKILLS

- R, Python, SQL, Bash, Git
- Statistical analysis
- IHC + Microscopy
- Slice electophysiology
- Biochemistry
- Mouse genetics

Extended skills: Cre-lox genetics, immunohistochemistry, confocal microscopy, electrophysiology, biochemistry, immunoprecipitation, SDS-PAGE, western blot, virus design, intracerebroventricular virus injection, primer design, PCR, 2-photon microscopy, glutamate uncaging, organotypic slice culture, biolistic transfection, genetically encoded fluorescent biosensors, experimental design, data analysis, hypothesis testing, cluster analysis, figure design, Adobe Photoshop, Adobe Illustrator, R, Python, SQL, Bash, Git

HONORS

2023 Vollum/NGP Paper of the Year
2023-2024 OHSU School of Medicine N.L. Tartar Trust Research Fellowship
2020-2023 Ruth L. Kirschstein National Research Service Award
2020 NSF GRFP Honorable Mention
ARCS Foundation Scholar
Phi Beta Kappa

This resume was made with the R package pagedown.

Last updated on 2026-11-05.

2018 | 2014

Staff Research Associate

Zito & Fioravante Labs, UC Davis — Davis, CA

- Characterization of genetically encoded fluorescent glutamate sensors using 2-photon glutamate uncaging.
- Elucidation of a novel protein signaling cascade in non-ionotropic NMDA receptor mediated LTD.

2014 | 2011

Undergraduate Research Assistant

Trimmer & Trainor Labs, UC Davis — Davis, CA

PROGRAMMING PROJECTS

2023

Calculating the o-index

UW Neurohackademy — Seattle, WA

A group project as part of UW Neurohackademy 2023. We wrote a program in Python that scraped open access publications for words indicating that data/code was publicly available. For a given author we then aggregated the "openness" of all of their published works to calculate an "o-index" for that given author, meant to be used as a standardized metric of openness. We wrote and deployed the program as a Shiny app on shinyapps.io.

Code available on GitHub

2022

Analyzing differentially regulated elements in response to heat shock across species

CSHL Programming for Biology — Cold Spring Harbor, NY

A group project as part of CSHL Programming for Biology 2022. We used publicly available datasets for experiments wherein different species were exposed to heat shock. We analyzed deferentially expressed genes, GO terms, and affected protein domains for all species to see which genes/terms/domains are changed similarly across species.

Code available on GitHub

2022

Deployed web application for randomly assigning application reviewers

Oregon Health & Science University — Portland, OR

A solo project where I was tasked with writing a python program that would take in a list of applicants to our graduate program and a list of reviewers and randomly assign reviewers to applicants such that each application is reviewed by n reviewers where n is defined by the user and ensures that faculty reviewers are the majority vote on each application. This application was born out of a need to ensure fair application review within our graduate program application process. The program was deployed as a Shiny app using Shiny for Python and has been used for hundreds of applications each year since the 2022 application cycle.

Code available on GitHub

2025 | 2021

Web scraping and GPS coordinate clustering for Portland food weeks

Portland, OR

A personal project born out of a love for the Portland food scene. Several times a year, Portland hosts these food weeks (pizza week, wing week, nacho week, etc.) where for one week only participating restaurants provide a dish on theme for the week and at a set price. Dozens of restaurants participate and I got tired of having to click through a bunch of links to see what each restaurant had to offer so I wrote a script in R to scrape the web and build out a spreadsheet with all of the offerings. I then deployed an R Shiny app where I could filter the table based on various qualities and vote on which ones I wanted to go to. Then I used GIS data to plot the restaurant locations and k-means cluster analysis to cluster restaurants based on GPS coordinates to identify which restaurants I might hit up in one go.

Write up available on my website, Full code available on GitHub

PUBLICATIONS

2026

Molecular assemblies and pharmacology of cerebellar GABAA receptors.

Sun C, Jahncke JN, Wright KW, Gouaux E. Proc Natl Acad Sci U S A. 2026;123(6):e2524504123. doi:10.1073/pnas.2524504123. doi: 10.1073/pnas.2524504123

2025

Distinct functional domains of Dystroglycan regulate inhibitory synapse formation and maintenance in cerebellar Purkinje cells.

Jahncke JN, Schnell E, Wright KW. Commun Biol. 2025; 8(1):878. doi: 10.1038/s42003-025-08323-1

2024

Tools for Cre-Mediated Conditional Deletion of Floxed Alleles from Developing Cerebellar Purkinje Cells.

Jahncke JN, Wright KW. eNeuro 22 May 2024, 11 (6) ENEURO.0149-24.2024; doi: 10.1523/ENEURO.0149-24.2024

2024

Inhibitory CCK+ basket synapse defects in mouse models of dystroglycanopathy.

Jahncke JN, Miller DS, Krush M, Schnell E, Wright KW. eLife. 2024; 12:RP87965. doi: 10.7554/eLife.87965.3

2023

An adhesion signaling axis involving Dystroglycan, β1-Integrin, and Cas adaptor proteins regulates the establishment of the cortical glial scaffold.

Wong W, Estep JA, Treptow AM, Rajabli N, Jahncke JN, Ubina T, Wright KM, Riccomagno MM. PLoS Biol. 2023 Aug 4;21(8):e3002212. doi: 10.1371/journal.pbio.3002212

2023

The many roles of dystroglycan in nervous system development and function.

Jahncke JN, Wright KW. Developmental Dynamics. 2023; 252(1): 61-80. doi: 10.1002/dvdy.516

2022

Shared and Distinct Functional Effects of Patient-Specific Tbr1 Mutations on Cortical Development.

Co M, Barnard RA, Jahncke JN, Grindstaff S, Fedorov LM, Adey AC, Wright KM, O'Roak BJ. Journal of Neuroscience 14 September 2022, 42 (37) 7166-7181; doi: 10.1523/JNEUROSCI.0409-22.2022

2020 **Molecular Mechanisms of Non-ionotropic NMDA Receptor Signaling in Dendritic Spine Shrinkage.**
Stein IS, Park DK, Flores JC, Jahncke JN, Zito K. Journal of Neuroscience 22 April 2020, JN-RM-0046-20; doi: 10.1523/JNEUROSCI.0046-20.2020

2017 **A Dual Role for the RhoGEF Ephexin5 in Regulation of Dendritic Spine Outgrowth.**
Hamilton AM, Lambert JT, Parajuli LK, Vivas O, Park DK, Stein IS, Jahncke JN, Greenberg ME, Margolis SS, Zito K. Molecular and Cellular Neuroscience 80 (2017): 66-7 doi: 10.1016/j.mcn.2017.02.001

TALKS

2024 **Go Nuts for Donuts: Molecular determinants of cerebellar GABAergic donut synapses**
Vollum Works In Progress — Portland, OR

2024 **A functional role for Dystroglycan at inhibitory synapses across multiple brain regions**
Dissertation Defense — Portland, OR

2023 **A conserved role for Dystroglycan at inhibitory synapses across multiple brain regions**
Vollum Works In Progress — Portland, OR

2023 **Inhibitory Basket Synapse Formation in Mouse Models of Dystroglycanopathy**
Vollum Works In Progress — Portland, OR

2021 **A functional role for Dystroglycan at inhibitory cerebellar synapses**
Vollum Works In Progress — Virtual

2020 **Dystroglycan: a scaffold for inhibitory synapses in cerebellar cortex**
Vollum Works In Progress — Virtual

ABSTRACTS

2026 **A conserved synaptic nanoarchitecture at a subset of cerebellar synapses lays the groundwork for strong GABAergic input onto Purkinje cell primary dendrites.**
Jahncke JN, Wright KM. CNCM Neural Circuits in Health & Disease
Irvine, CA

2025 **Distinct domains of Dystroglycan regulate GABAergic synapse formation and function in cerebellar Purkinje cells.**
Jahncke JN, Schnell E, Wright KM. Inhibition in the CNS GRC
Newry, ME

2023 **Purkinje neuron Dystroglycan is required for inhibitory synapse formation and maintenance.**
Jahncke JN, Schnell E, Wright KM. OHSU NGP Retreat
Government Camp, OR

2022 **Dystrogycan is necessary for the formation of functional inhibitory CCK+/CB1R+ basket synapses in hippocampus.**
Jahncke JN, Miller DS, Schnell E, Wright KM. CSHL Molecular Mechanisms of Neuronal Connectivity
Cold Spring Harbor, NY

2020 **Purkinje cell Dystroglycan may promote the function of inhibitory basket synapses in cerebellar cortex.**
Jahncke JN, Schnell E, Wright KM. CSHL Molecular Mechanisms of Neuronal Connectivity
Cold Spring Harbor, NY

2019 **Non ionotropic NMDA receptor signaling mechanisms driving dendritic spine plasticity.**
Stein IS, Park DK, Jahncke JN, Zito K. Society for Neuroscience
Chicago, IL

2018 **Genetically-encoded glutamate sensors for monitoring large populations of individual synapses in vivo.**
Mizuno GO, Dong C, Jahncke JN, Natan R, Papadopoulos S, Jaffe D, Lambert JT, Ji N, Zito K, Tian . Fourth Annual Brain Initiative Investigators Meeting
Bethesda, MD

TEACHING EXPERIENCE

2025 | 2026 **MBL Analytical and Quantitative Light Microscopy**
Teaching assistant at the Marine Biological Laboratory AQLM course.
Woods Hole, MA

2023 | 2025 **OHSU NGP Bootcamp: Model Organisms**
Hands on dissection of mouse and sheep brains to familiarize incoming graduate students with practical neuroanatomy of model system organisms.
Portland, OR

2023 | 2026 **Rigor and Reproducibility in Light Microscopy**
Lecture on best practices in light microscopy, aimed at incoming neuroscience graduate students.
Portland, OR

2023 | 2025 **OHSU MD Foundations of Medicine: Nervous System and Function**
Instruction of first year medical students learning about brain structure and function. Taught human cerebellum and brain stem anatomy in a hands on lab working with donated specimen.
Portland, OR

2021 **OHSU NGP Bootcamp: Patch Clamp Electrophysiology**
Hands on patch clamp instruction. Portland, OR

2021, 2019 **Graduate Level TA Positions**
Biostatistics (CONJ 620) Winter 2021, OHSU Portland, OR
Cellular Neurophysiology (NEUS 624) Fall 2019, OHSU

2018 **MBL Neurobiology, Imaging Section**
Teaching assistant at the Marine Biological Laboratory Neurobiology Course for the imaging section.
Woods Hole, MA

VOLUNTEER & MENTORSHIP EXPERIENCE

2022 | 2026 **Neuroscience Graduate Program Rotation Student Mentor**
Mentored students through their 8-week laboratory rotation, overseeing their laboratory work, data analysis, and data presentation. In total I have mentored six rotation students.
Portland, OR

2024 **Undergraduate Researcher Mentor**
Mentored an undergraduate research intern as part of the two month OHSU NGP Summer Intern Program. Oversaw their laboratory work, data analysis, oral presentation, and poster presentation.
Portland, OR

2022 | 2020 **Alliance for Visible Diversity (AVDS)**
Communications Committee Member — Portland, OR

2022 | 2019 **OHSU Graduate School Organization (GSO) Peer Mentoring Program**
Peer Mentor — Portland, OR

2020 | 2019 **Women In Science PDX (WIS)**
STEMpowerment Committee Member — Portland, OR

2019 | 2018 **NGP Alumni Invite Committee**
Committee Member — Portland, OR

2018 | 2015 **Brain Awareness Week - Davis Chapter**
Event Organizer, Volunteer — Davis, CA

2015 **UC Davis Neurobiology, Physiology, & Behavior Club**
Guest Speaker — Davis, CA

2018 | 2014 **Undergraduate Researcher Mentor**
Mentored two undergraduates in the lab setting, overseeing their independent research projects and providing support through poster presentations, fellowship applications, and career development.
Davis, CA